

Discharge Summary

Dear Colleagues,

Mr. Klaus-Dieter Richter, DOB 12/22/1954, was admitted on 01/02/2025 to initiate VCD induction chemotherapy for newly diagnosed symptomatic multiple myeloma. Mr. Richter presented 3 months ago with progressive low back pain, fatigue, and an incidentally elevated total protein. IgG kappa paraprotein 42 g/L. Bone marrow biopsy: 65% plasma cell infiltration. Whole-body CT: multiple lytic lesions in thoracic and lumbar vertebrae and pelvis, with a T8 pathological compression fracture. Serum calcium normal. ISS stage III.

Primary Diagnosis: Multiple myeloma (C90.0) — Tumor staging: T3 N1 M1

Laboratory values on admission:

Parameter	Value	Reference
HbA1c	6.7%	< 7.0%
Total protein	9.8 g/dL	6.0–8.3 g/dL
Calcium	2.8 mmol/L	2.1–2.6 mmol/L
Creatinine	1.9 mg/dL	0.7–1.2 mg/dL

Past Medical History: Type 2 diabetes since 2010 on sitagliptin 100 mg and metformin 1000 mg twice daily (HbA1c 6.7%). Mild hypertension on lisinopril 5 mg. Laparoscopic cholecystectomy 2008. Retired bank manager, non-smoker. Wife is primary carer and attended all key discussions.

Pre-Treatment Assessment: Echo: EF 60%, no amyloid features, no pericardial effusion. ECG: normal sinus rhythm. eGFR 52 mL/min — no bortezomib dose adjustment required. Spine MRI confirmed T8 fracture, no cord compromise.

Hospital Course: Cycle 1 of VCD (bortezomib subcutaneous days 1, 4, 8, 11; cyclophosphamide days 1, 8; dexamethasone 40 mg on days 1, 2, 4, 5, 8, 9, 11, 12) was administered. Aciclovir 400 mg twice daily for zoster prophylaxis.

Zoledronic acid 4 mg IV day 3 for bone disease. Blood glucose monitoring intensified on dexamethasone days — glucose peaked 268 mg/dL day 1, managed with insulin sliding scale. Sitagliptin held on dexamethasone days. Grade 1 peripheral neuropathy noted day 10 — to be monitored closely. Pain management with regular paracetamol and transdermal buprenorphine for vertebral fracture pain effective.

Pain and Physiotherapy: Physiotherapy assessed on days 2 and 6. Mobility limited by vertebral fractures. Walking frame provided, gait training initiated. OT assessed home environment and recommended bathroom adaptations. Community physiotherapy referral made.

Medication at Discharge: Sitagliptin 100 mg (held on dexamethasone days), metformin 1000 mg twice daily, lisinopril 5 mg, aciclovir 400 mg twice daily, zoledronic acid next dose at outpatient visit, omeprazole 20 mg, transdermal buprenorphine 5 mcg/h.

Follow-up: Hematology in 3 weeks for pre-cycle 2 assessment. Paraprotein and CBC after cycle 2. Response CT and bone marrow biopsy after cycle 4.

Pain Management for Vertebral Disease:

Mr. Richter's back pain from myeloma bone disease and the T8 pathological compression fracture was severe on admission (7/10 at rest). The haematology team consulted the palliative care pain specialist who recommended multimodal analgesia within renal function constraints (eGFR 52 mL/min). Regular paracetamol 1 g four times daily was established as the foundation. NSAIDs were avoided given renal function and myeloma-related renal risk. Transdermal buprenorphine 5 mcg/h was added on day 3 following inadequate response. Pain scores improved to 3/10 by day 5.

Physiotherapy, Occupational Therapy, and Home Adaptations:

The physiotherapy team assessed Mr. Richter on days 2 and 6. Mobility was significantly limited by the vertebral fracture, requiring a walking frame for distances greater than 10 metres. Gait training was initiated and graduated exercise tolerance was assessed. The occupational therapist conducted a home environment assessment and recommended: installation of a raised toilet seat,

addition of bathroom grab rails, and replacement of the low armchair with a higher-seated chair to reduce spinal loading during sit-to-stand transfers. The wife confirmed she would arrange for these adaptations before the patient's discharge. A community physiotherapy referral was made for continuation of the rehabilitation programme at home.

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